

# Retrieval-Augmented Generation

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Lara J. Martin (she/they)

<https://laramartin.net/neurosymbolic-text-gen/>

*Slides adapted from an ACL 2023 Tutorial by Akari Asai, Sewon Min, Zexuan Zhong, & Dr. Danqi Chen*

# Learning Objectives

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Define the Story Cloze Test and determine its place in guided story generation

Understand the reasons why RAG was created

Explore how the retrieval component interacts with the LLM in RAG

Extract implementation details from papers and find different ways RAG is implemented

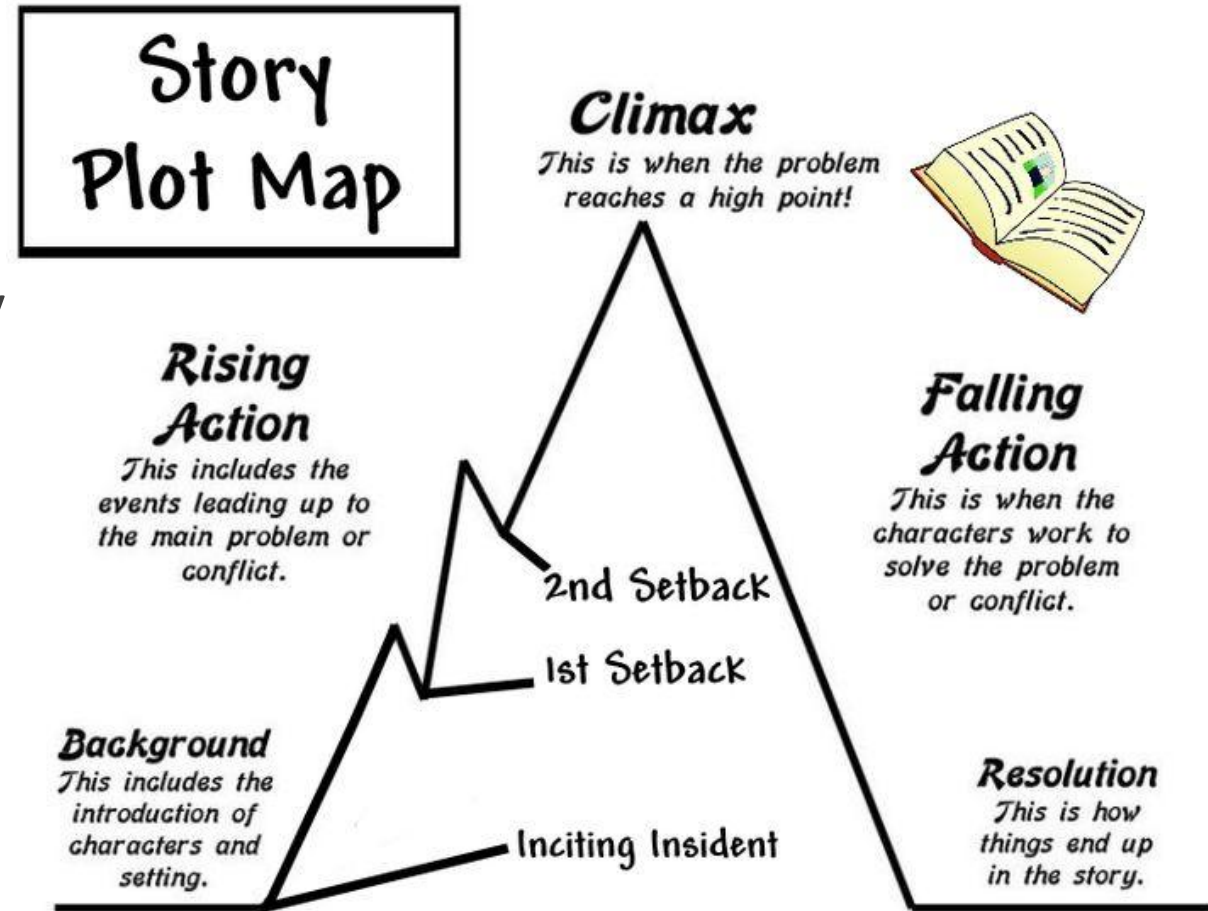
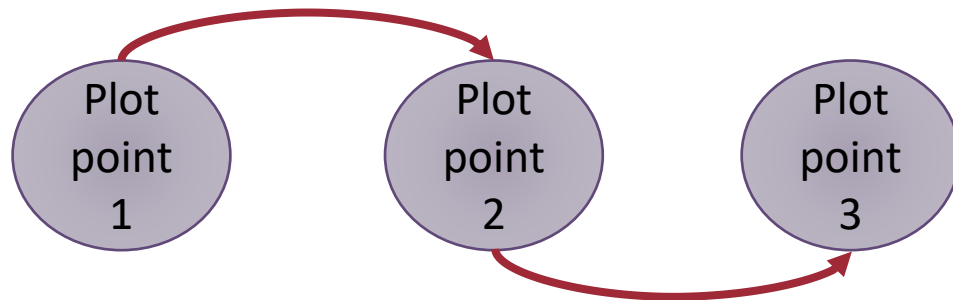
Compare plot-guided generation to retrieval-augmented generation for stories

# Review:

## Scripts, Procedures, and Plots...oh my!

Schank & Abelson believe that everyone has scripts in their heads built from common experiences

Authors often plan out **plots** before they write stories



<https://i.pinimg.com/736x/57/f7/03/57f703afc709080bddc2c3cfed8dd061.jpg>

# Review:

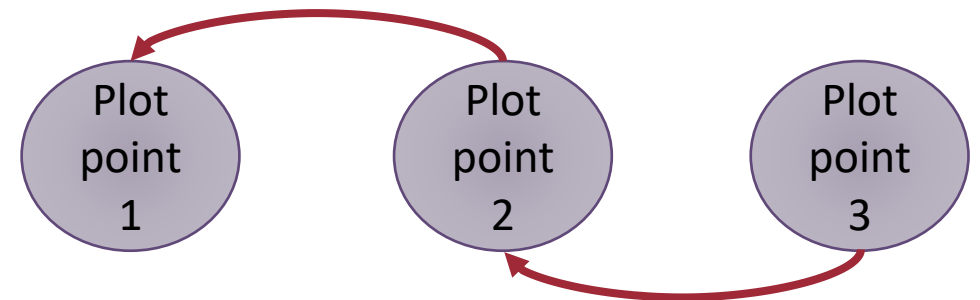
## Scripts, Procedures, and Plots...oh my!

---

Schank & Abelson believe that everyone has scripts in their heads built from common experiences

Authors often plan out plots before they write stories

Stories that aren't planned out either have to “**reincorporate**”[1] ideas or the stories feel unfinished



[1] The idea of *reincorporation* is explored in the book [Impro by Keith Johnstone](#)

# Review: Ways of Extracting Plot Points

Most salient keywords

Event representations

Verb-Noun Sets

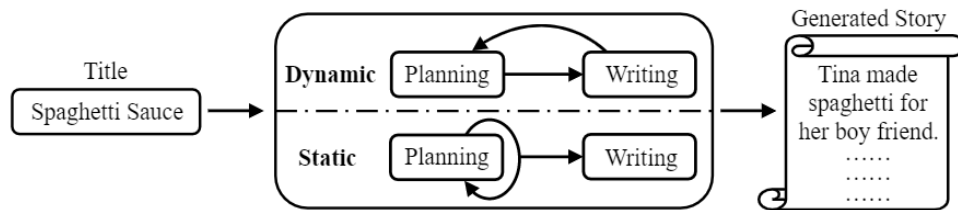
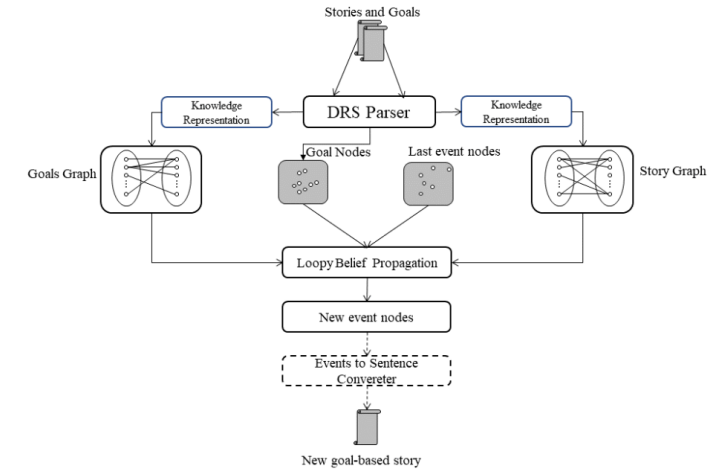
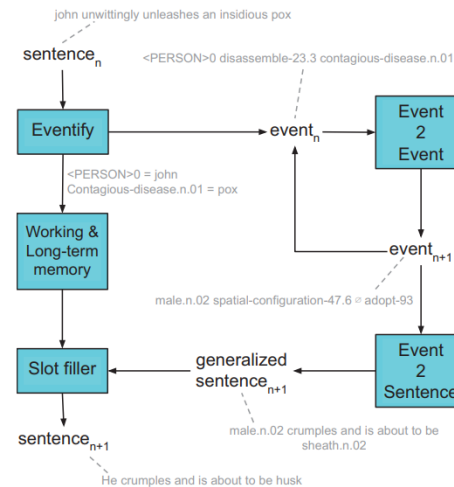


Figure 1: An overview of our system.



# Review: Generating with Plot Points

Co-generated vs conditioned (prompted) with plot

Generate event & then translate to natural language

Graph algorithms (Loopy Belief Propagation)

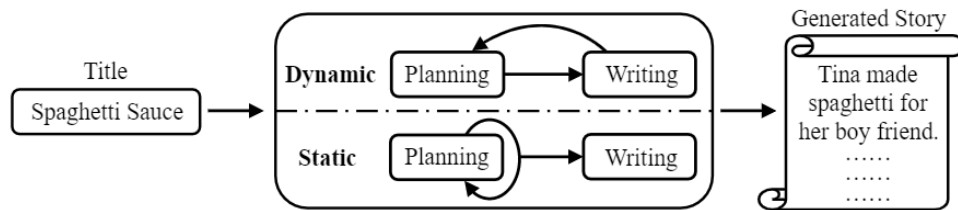
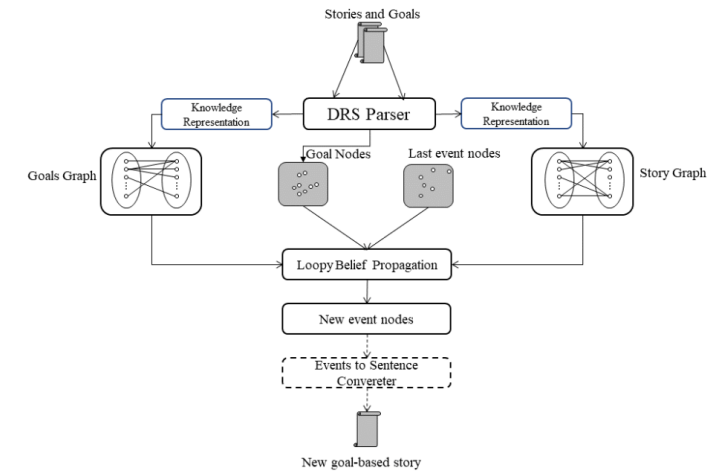
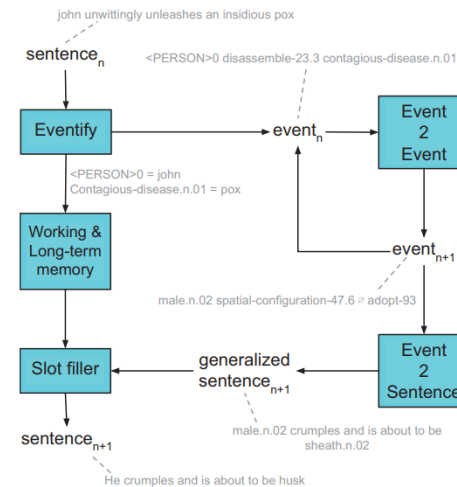


Figure 1: An overview of our system.



# The Story Cloze Test

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# What is a Cloze Test?

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- Something is removed from a text; try to guess what's missing
- Used for reading comprehension, grammar, etc. (with humans)

# Narrative Cloze Test

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Evaluate “event relatedness”

Find which events could be missing from a narrative chain

Uses verbs only

# Narrative Cloze Test

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**Known events:**

(pleaded subj), (admits subj), (convicted obj)

**Likely Events:**

|               |      |              |      |
|---------------|------|--------------|------|
| sentenced obj | 0.89 | indicted obj | 0.74 |
| paroled obj   | 0.76 | fined obj    | 0.73 |
| fired obj     | 0.75 | denied subj  | 0.73 |

X pleaded \_

X admits \_

\_ convicted X

Figure 1: Three narrative events and the six most likely events to include in the same chain.

# Finish the story

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Gina was worried the cookie dough in the tube would be gross.

She was very happy to find she was wrong.

The cookies from the tube were as good as from scratch.

Gina intended to only eat 2 cookies and save the rest.

A. Gina liked the cookies so much she ate them all in one sitting.



B. Gina gave the cookies away at her church.

# Story Cloze Test

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Predict/select the most likely story \*ending\*

- Given the first 4 sentences of the story

Full sentences

Multiple choice evaluation

## An RNN-based Binary Classifier for the Story Cloze Test

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## Toward Better Storylines with Sentence-Level Language Models

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### Abstract

We propose a sentence-level language model which selects the next sentence in a story from a finite set of fluent alternatives. Since it does not need to model fluency, the sentence-level language model can focus on longer range dependencies, which are crucial for multi-sentence coherence. Rather than dealing with individual words, our method treats the story so far as a list of pre-trained sentence embeddings and predicts an embedding for the next sentence, which is more efficient than predicting word embeddings. Notably this allows us to consider a large number of candidates for the next sentence during training. We demonstrate the effectiveness of our approach with state-of-the-art accuracy on the unsupervised Story Cloze task and with promising results on larger-scale next sentence prediction tasks.

quence of images  
roles (Liu et al.)

Our work is  
than considering  
pose a model  
of context and  
a large set of  
age pre-trained  
2019) to build  
Given the em  
of the story, c  
embedding of

This task is  
dependencies  
words, which  
our model on  
candidate sen  
tuation to th  
and time to le

## Tackling the Story Ending Biases in The Story Cloze Test

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<sup>1</sup> University of Rochester, <sup>2</sup> Institute for Human and Machine Cognition, <sup>3</sup> Verneek.ai <sup>4</sup> Elemental Cognition  
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### Abstract

The Story Cloze Test (SCT) is a recent framework for evaluating story comprehension and script learning. There have been a variety of models tackling the SCT so far. Although the original goal behind the SCT was to require systems to perform deep language understanding and commonsense reasoning for successful narrative understanding, some recent models could perform significantly better than the initial baselines by leveraging human-authorship biases discovered in the SCT dataset. In order to shed some

## Enhanced Story Representation by ConceptNet for Predicting Story Endings

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### ABSTRACT

Predicting endings for machine commonsense representation of the story (Pre-trained language) in this task by exploiting dataset, instead of "u we propose to improve the sentences to latent relationship between enhanced sentence regression models, makes the popular Story Cloze data.

### CCS CONCEPTS

IEEE/ACM TRANSACTIONS ON AUDIO, SPEECH, AND LANGUAGE PROCESSING, VOL. 27, NO. 4, APRIL 2019

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## Story Ending Selection by Finding Hints From Pairwise Candidate Endings

Mantong Zhou<sup>1</sup>, Minlie Huang<sup>2</sup>, and Xiaoyan Zhu

strong indicator  
Story Cloze Test  
ending candidate  
ending methods  
d that operate  
text, therefore  
te endings can  
which misleads  
ress this issue,  
sion by utiliz-  
two candidate  
feature vector  
id then refines  
the difference  
se feature vec-  
is regarded as  
approach can  
comprehension.  
story compre-



Fig. 1. Evidence bias issue: both a wrong ending (in red) and a correct ending (in green) can obtain sufficient evidence from the story context.

important linkages between a story context and a candidate ending. They suffer from the issue of **evidence bias**: both the wrong and correct endings can obtain sufficient support from the story context. As illustrated in Fig. 1, the wrong ending (in red) and the correct ending (in green) can be supported by the red-colored evidence and the green-colored evidence in the story context, respectively. Thus, it is difficult for matching-based models to distinguish such cases. The situation is not rare because both correct and wrong endings are written to fit the world of a story.

# Extending your thinking

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The Story Cloze Test was created for evaluating systems' performance on understanding stories.

How could you use it instead for *generation*?

# Retrieval-Augmented Generation

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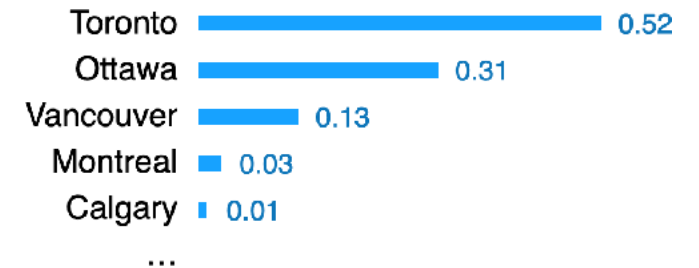
# Retrieval-based language models (LMs)

## Retrieval-based LMs = Retrieval + LMs

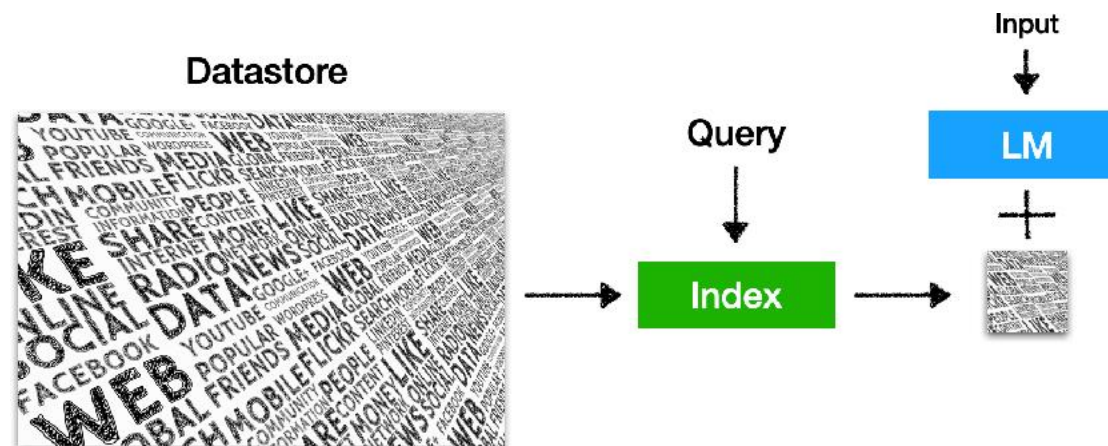
- It is a **language model**  $P(x_n | x_1, x_2, \dots, x_{n-1})$

The capital city of Ontario is \_\_\_\_

(can be broadly extended to masked language models or encoder-decoder models)



- It retrieves from an **external datastore** (at least during inference time)



# Retrieval for knowledge-intensive NLP tasks

**Representative tasks:** open-domain QA, fact checking, entity linking, ..

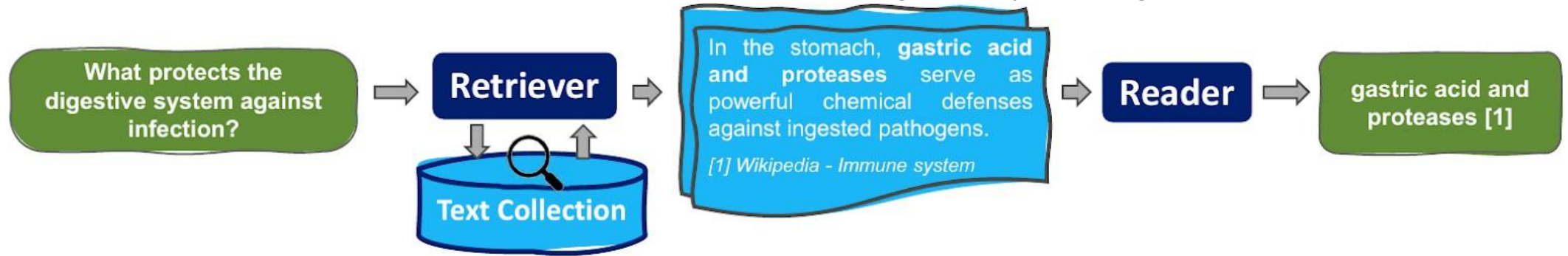


Image: <http://ai.stanford.edu/blog/retrieval-based-NLP/>

Drives a lot of research on better algorithms for **dense retrieval**, e.g., **DPR** (Karpukhin et al., 2020), **CoBERT** (Khattab and Zaharia, 2020), **ANCE** (Xiong et al., 2021), **Contriever** (Izacard et al., 2022), ...

Why retrieval LMs?

# Why retrieval-based LMs?



List 5 important papers authored by Geoffrey Hinton



Geoffrey Hinton is a renowned computer scientist ...  
Here are five important papers authored by him:

1. **"Learning Internal Representations by Error Propagation"** (with **D. E. Rumelhart** and **R. J. Williams**) - This paper, published in **1986**, ..



2. **"Deep Boltzmann Machines"** (with **R. Salakhutdinov**) - Published in **2009**, ..



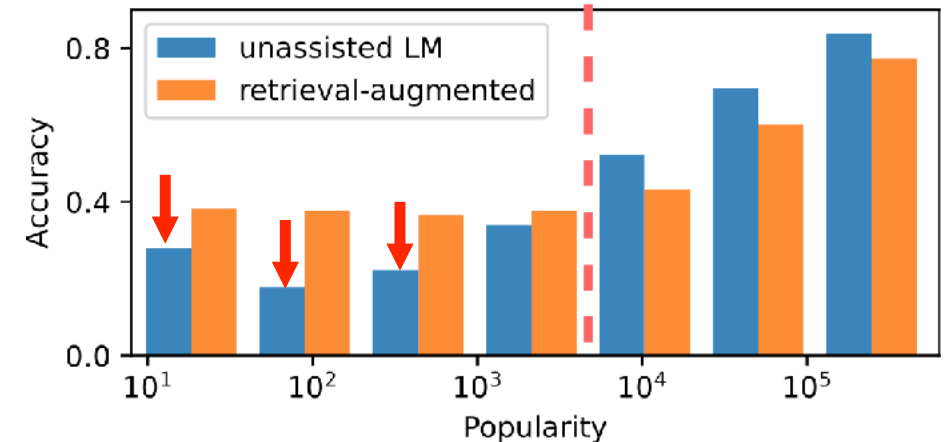
4. **"Deep Learning"** (with Y. Bengio and A. Courville) -  
Published as a book in **2016**,...



5. **"Attention Is All You Need"** (with V. Vaswani, N. Shazeer, et al.) - Published in **2017**, this paper introduced the Transformer model,...



What is Kathy Saltzman's occupation?



(Mallen et al., 2023)

GPT-3 davinci-003: 20%-30% accuracy

# Why retrieval-based LMs?



Who is the CEO of Twitter?



As of my **knowledge cutoff in September 2021**, the CEO of Twitter is **Jack Dorsey**....



Who is the CEO of Twitter?



About 1,090,000,000 results (0.45 seconds)

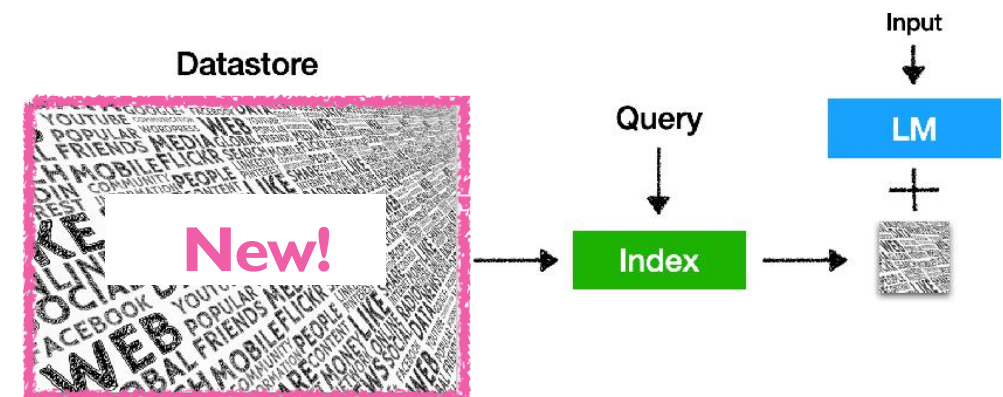
Twitter / CEO

Linda Yaccarino

Jun 5, 2023–



- Existing **knowledge editing** methods are still NOT scalable (**active research!**)
- The datastore can be easily **updated** and **expanded** - even without retraining!



# Actually Google AI gets it wrong too...

## ✦ AI Overview

There is no CEO of Twitter (now X); Elon Musk is still the owner, but he stepped down as CEO in July 2025, and Linda Yaccarino also stepped down from the position at the same time. Musk now serves as the company's owner, chairman, and CTO. [🔗](#)

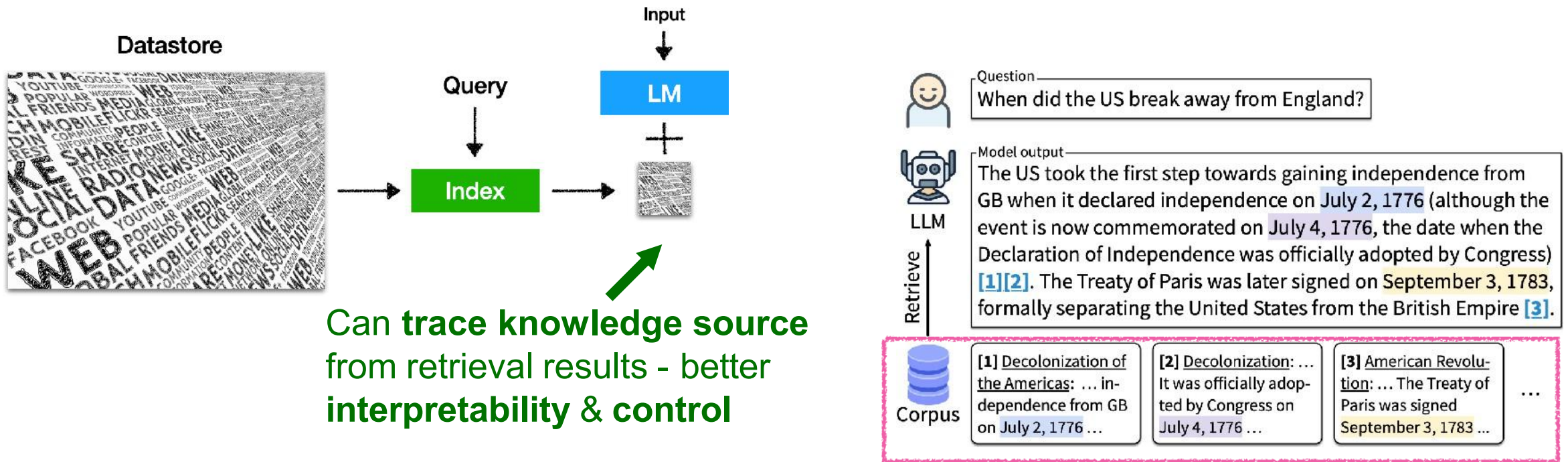


## A Brief History of the CEO role

- **Elon Musk:** Musk was the CEO of X (formerly Twitter) immediately after acquiring the company in October 2022. [🔗](#)
- **Linda Yaccarino:** Yaccarino replaced Musk as CEO on June 5, 2023, but she resigned in July 2025. [🔗](#)
- **Elon Musk's Current Role:** Musk has remained the owner and is now serving as the chairman and chief technology officer for X. [🔗](#)

# Why retrieval-based LMs?

Generating text with citations



Can trace knowledge source from retrieval results - better interpretability & control

(Nakano et al. 2021; Menick et al., 2022; Gao et al., 2023)

# Why retrieval-based LMs?

 Create an itinerary for exploring Toronto over a span of two days.  



Toronto is a vibrant and diverse city with many attractions and activities to suit different tastes and preferences. Here are some suggestions for creating an itinerary for exploring Toronto over two days:

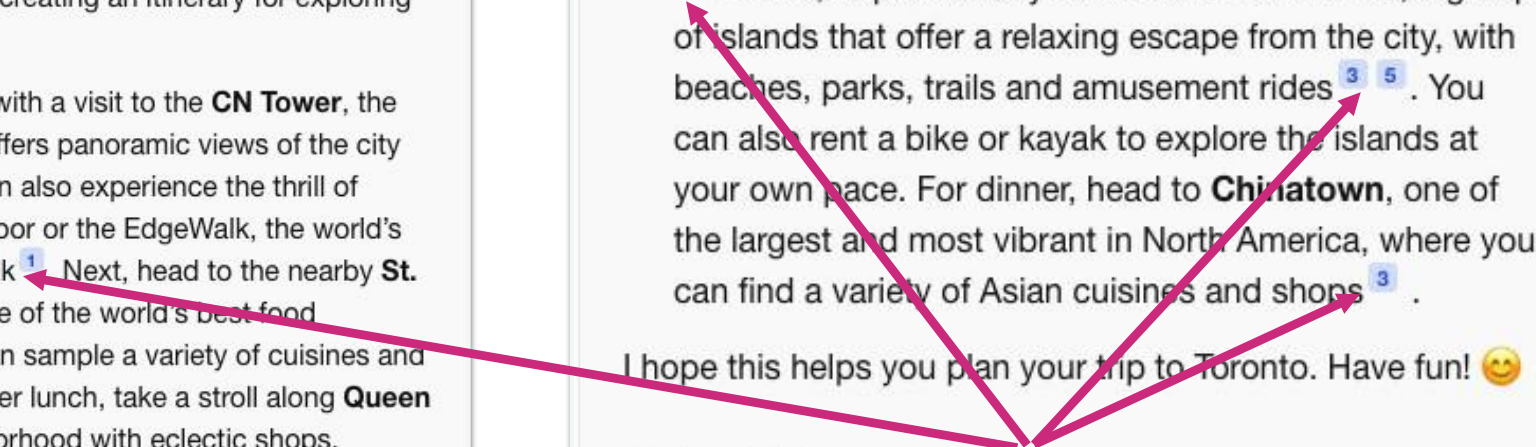
- **Day 1:** Start your day with a visit to the **CN Tower**, the iconic landmark that offers panoramic views of the city and beyond <sup>1</sup>. You can also experience the thrill of walking on the glass floor or the EdgeWalk, the world's highest hands-free walk <sup>1</sup>. Next, head to the nearby **St. Lawrence Market**, one of the world's best food markets, where you can sample a variety of cuisines and local specialties <sup>2</sup>. After lunch, take a stroll along **Queen West**, a trendy neighborhood with eclectic shops, galleries, cafes and street art <sup>3</sup>. In the evening, enjoy a show at one of the many theaters or comedy clubs in the **Entertainment District**, or catch a game at the **Scotiabank Arena** if you're a sports fan.

- **Day 2:** Explore the history and culture of Toronto at the **Royal Ontario Museum**, one of the largest and most comprehensive museums in North America, featuring exhibits on art, natural history, world cultures and more <sup>4</sup>. Then, hop on a ferry to the **Toronto Islands**, a group of islands that offer a relaxing escape from the city, with beaches, parks, trails and amusement rides <sup>3</sup> <sup>5</sup>. You can also rent a bike or kayak to explore the islands at your own pace. For dinner, head to **Chinatown**, one of the largest and most vibrant in North America, where you can find a variety of Asian cuisines and shops <sup>3</sup>.

I hope this helps you plan your trip to Toronto. Have fun! 😊

**Learn more:**

[1. cntower.ca](#) [2. travel.usnews.com](#) [3. bing.com](#)  
[4. rom.on.ca](#) [5. tripadvisor.com](#)

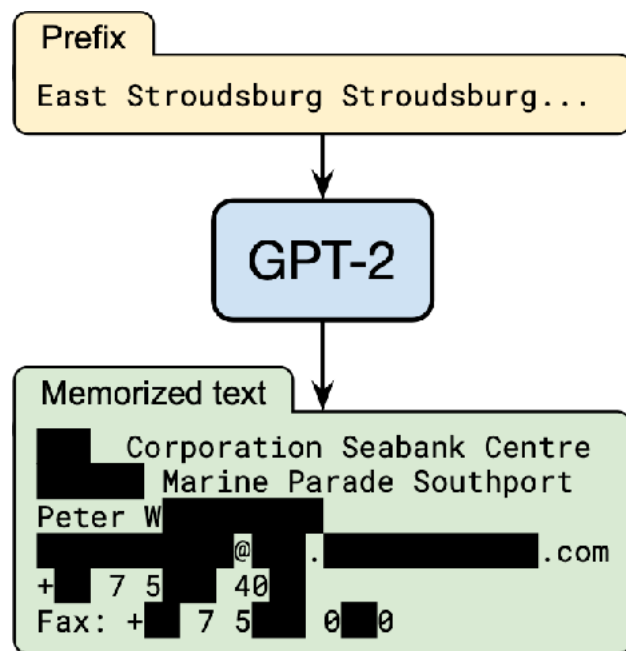


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RETRIEVAL-AUGMENTED GENERATION

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# Why retrieval-based LMs?

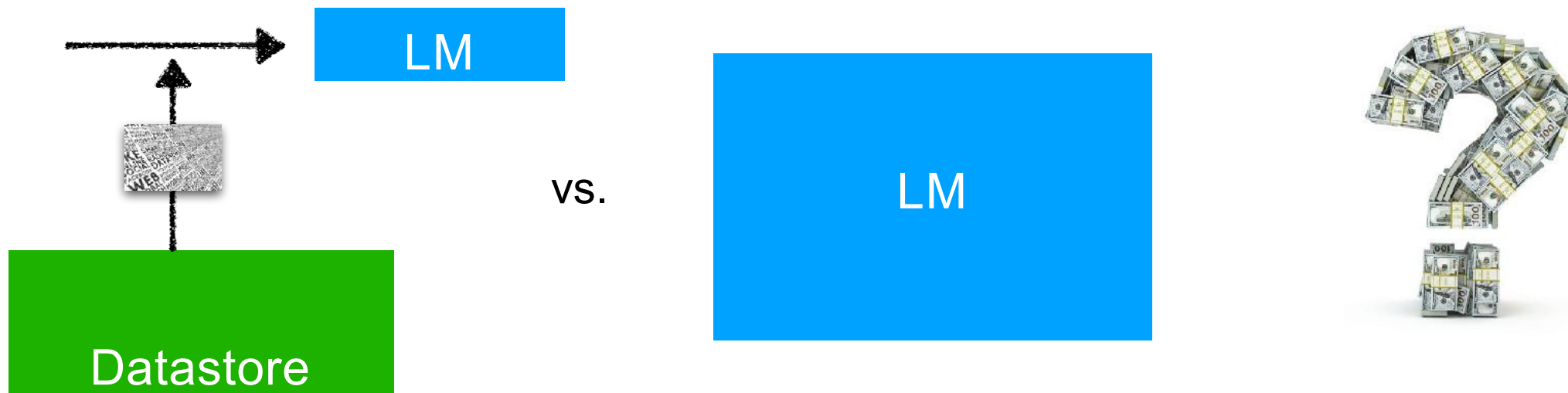


| Category                                                   | Count     |
|------------------------------------------------------------|-----------|
| US and international news                                  | 109       |
| Log files and error reports                                | 79        |
| License, terms of use, copyright notices                   | 54        |
| Lists of named items (games, countries, etc.)              | 54        |
| Forum or Wiki entry                                        | 53        |
| Valid URLs                                                 | 50        |
| <b>Named individuals (non-news samples only)</b>           | <b>46</b> |
| Promotional content (products, subscriptions, etc.)        | 45        |
| High entropy (UUIDs, base64 data)                          | 35        |
| <b>Contact info (address, email, phone, twitter, etc.)</b> | <b>32</b> |
| Code                                                       | 31        |
| Configuration files                                        | 30        |
| Religious texts                                            | 25        |
| Pseudonyms                                                 | 15        |
| Donald Trump tweets and quotes                             | 12        |
| Web forms (menu items, instructions, etc.)                 | 11        |
| Tech news                                                  | 11        |
| Lists of numbers (dates, sequences, etc.)                  | 10        |

Individualization on private data by storing it in the datastore

LLMs are \*large\* and expensive to train and run

# Why retrieval-based LMs?



**Long-term goal:** can we possibly reduce the **training** and **inference costs**, and scale down the size of LLMs?

e.g., RETRO (Borgeaud et al., 2021): “obtains comparable performance to GPT-3 on the Pile, despite using **25x fewer parameters**”

# A Retrieval-based LM: Definition

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A language model (LM) that uses  
**an external datastore at test time**

# Typical LMs



The capital city of Ontario is **Toronto**



LM

Training time

The capital city of Ontario is \_\_\_\_\_



LM

Test time / Inference

# Retrieval-based LMs



The capital city of Ontario is **Toronto**



LM

# Training time



The capital city of Ontario is \_\_\_\_\_



LM

## Test time / Inference

[illegible]

```

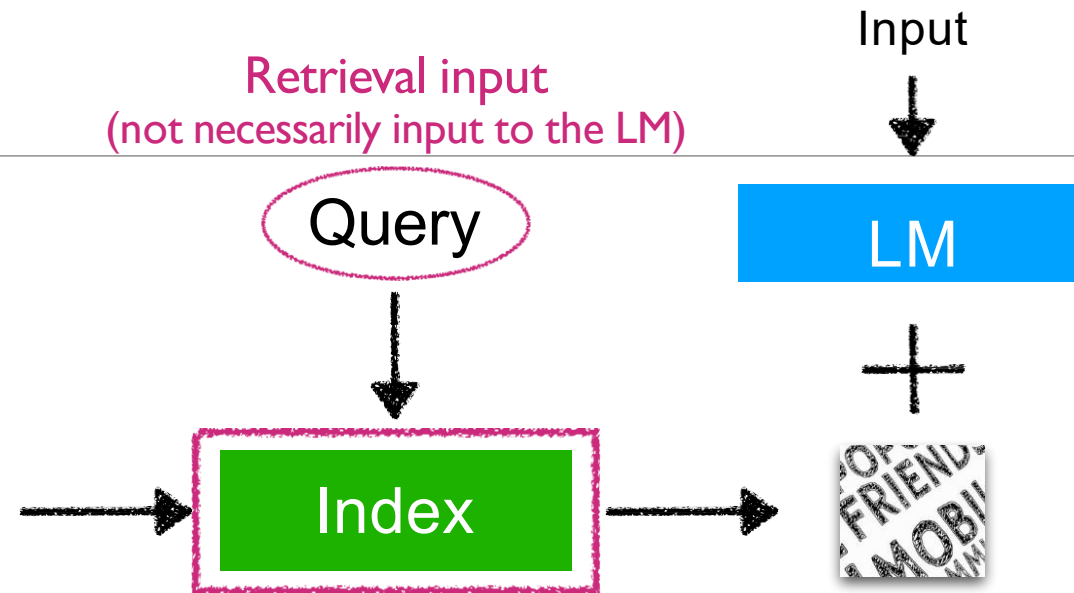
graph TD
    Input --> LM[LM]
    Query --> LM
    Query --> Index[Index]
    Index --> Plus[+]
    Plus --> Output[Output]
  
```



# Inference: Index



Datastore



Find a small subset of elements in a datastore that are the most similar to the query

# Inference: Index

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Goal: find a small subset of elements in a datastore that are the most similar to the query

**sim:** a similarity score between two pieces of text

# Inference: Index

Goal: find a small subset of elements in a datastore that are the most similar to the query

**sim**: a similarity score between two pieces of text

Example  $\text{sim}(i, j) = \text{tf}_{i,j} \times \log \frac{N}{\text{df}_i}$

$\text{tf}_{i,j}$ : # of occurrences of in  $j$

$N$ : # of total docs

$\text{df}_i$ : # of docs containing

Remember cosine similarity from our discussion of word embeddings

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Example  $\text{sim}(i, j) = \text{Encoder}(i) \cdot \text{Encoder}(j)$

Encoder( $i$ ): Maps the text into an -dimensional vector

Encoder( $j$ ): Maps the text into an -dimensional vector

# Inference: Index

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Goal: find a small subset of elements in a datastore that are the most similar to the query

**sim**: a similarity score between two pieces of text

Can be a totally separate research area on how to do this fast & accurate

**Index**: given  $q$ , return  $\text{argTop-}k_{d \in \mathcal{D}} \text{sim}(q, d)$  through fast nearest neighbor search  
 $k$  elements from a datastore

# Software: FAISS, Distributed FAISS, SCaNN, etc...

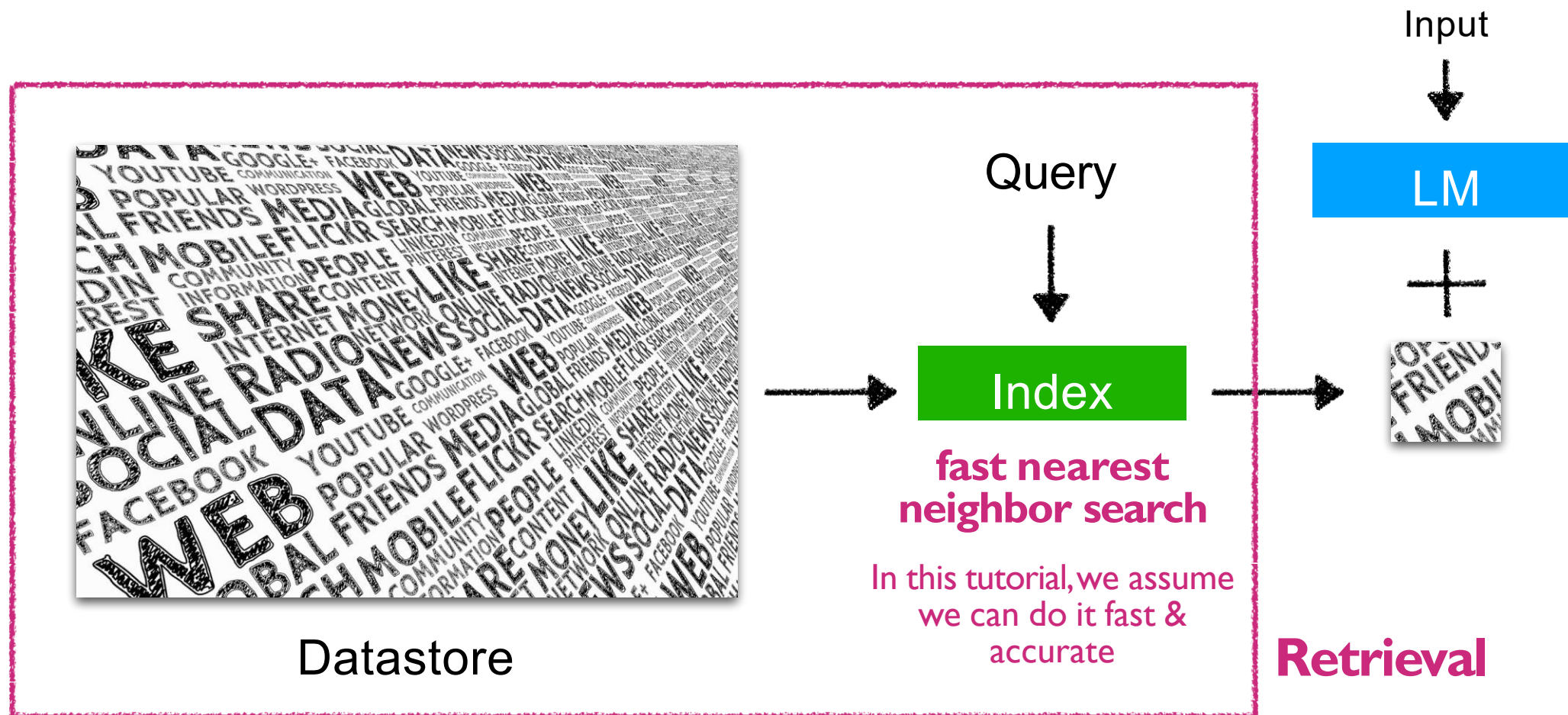
| Method                                                   | Class name              | index_factory            | Main parameters                                        | Bytes/vector                                  | Exhaustive | Comments                                                                                                                   |
|----------------------------------------------------------|-------------------------|--------------------------|--------------------------------------------------------|-----------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------|
| Exact Search for L2                                      | IndexFlatL2             | "Flat"                   | d                                                      | 4*d                                           | yes        | brute-force                                                                                                                |
| Exact Search for Inner Product                           | IndexFlatIP             | "Flat"                   | d                                                      | 4*d                                           | yes        | also for cosine (normalize vectors beforehand)                                                                             |
| Hierarchical Navigable Small World graph exploration     | IndexHNSWFlat           | "HNSW,Flat"              | d, M                                                   | $4*d + x * M * 2 * 4$                         | no         |                                                                                                                            |
| Inverted file with exact post-verification               | IndexIVFFlat            | "IVFx,Flat"              | quantizer, d, nlists, metric                           | 4*d + 8                                       | no         | Takes another index to assign vectors to inverted lists. The 8 additional bytes are the vector id that needs to be stored. |
| Locality-Sensitive Hashing (binary flat index)           | IndexLSH                | -                        | d, nbits                                               | ceil(nbbits/8)                                | yes        | optimized by using random rotation instead of random projections                                                           |
| Scalar quantizer (SQ) in flat mode                       | IndexScalarQuantizer    | "SQ8"                    | d                                                      | d                                             | yes        | 4 and 6 bits per component are also implemented.                                                                           |
| Product quantizer (PQ) in flat mode                      | IndexPQ                 | "PQx",<br>"PQ"M"x"nbbits | d, M, nbits                                            | ceil(M * nbbits / 8)                          | yes        |                                                                                                                            |
| IVF and scalar quantizer                                 | IndexIVFScalarQuantizer | "IVFx,SQ4"<br>"IVFx,SQ8" | quantizer, d, nlists, qtype                            | SQfp16: 2 * d + 8, SQ8: d + 8 or SQ4: d/2 + 8 | no         | Same as the IndexScalarQuantizer                                                                                           |
| IVFADC (coarse quantizer+PQ on residuals)                | IndexIVFPQ              | "IVFx,PQ"y"x"nbbits      | quantizer, d, nlists, M, nbits                         | ceil(M * nbbits/8)+8                          | no         |                                                                                                                            |
| IVFADC+R (same as IVFADC with re-ranking based on codes) | IndexIVFPQR             | "IVFx,PQy+z"             | quantizer, d, nlists, M, nbits, M_refine, nbits_refine | M+M_refine+8                                  | no         |                                                                                                                            |

Exact Search

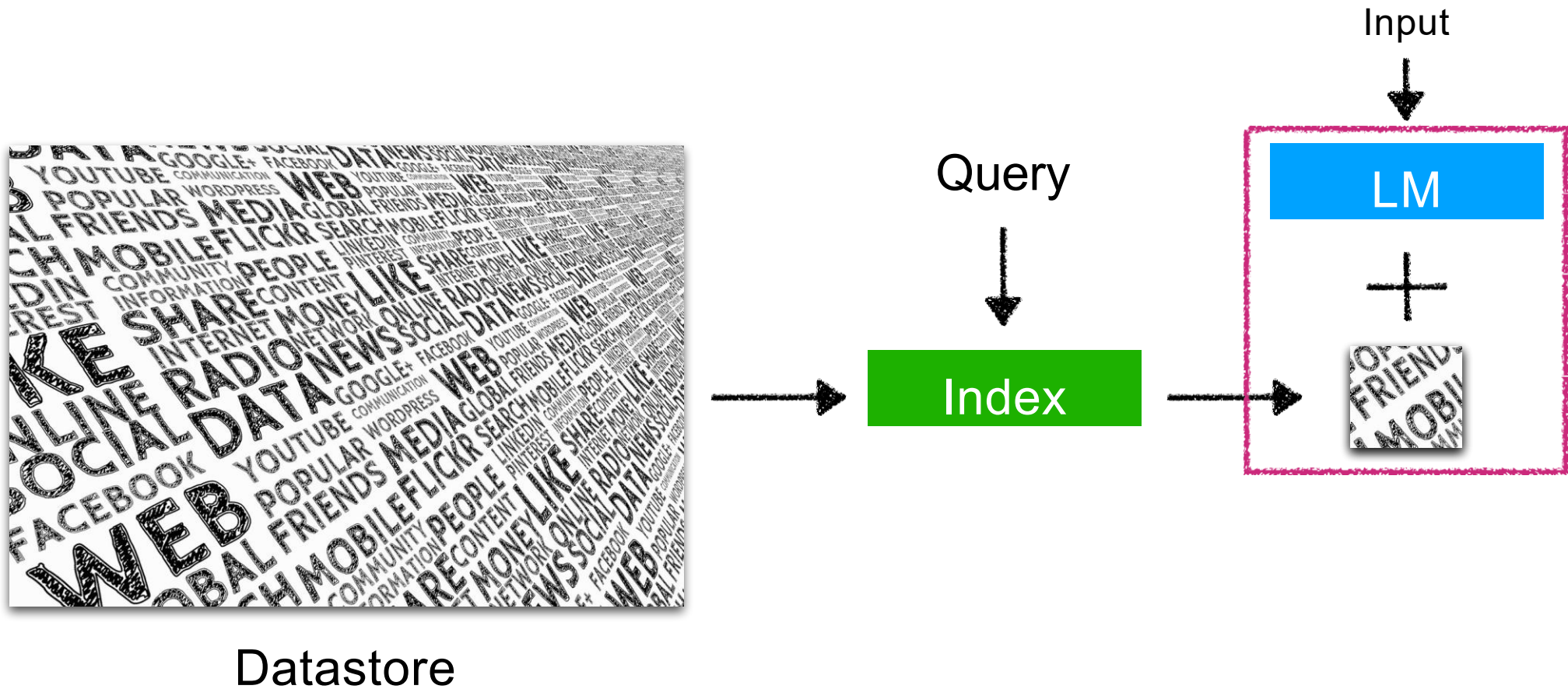
Approximate Search  
(Relatively easy to scale to ~1B elements)

More info: <https://github.com/facebookresearch/faiss/wiki>

# Inference: Search



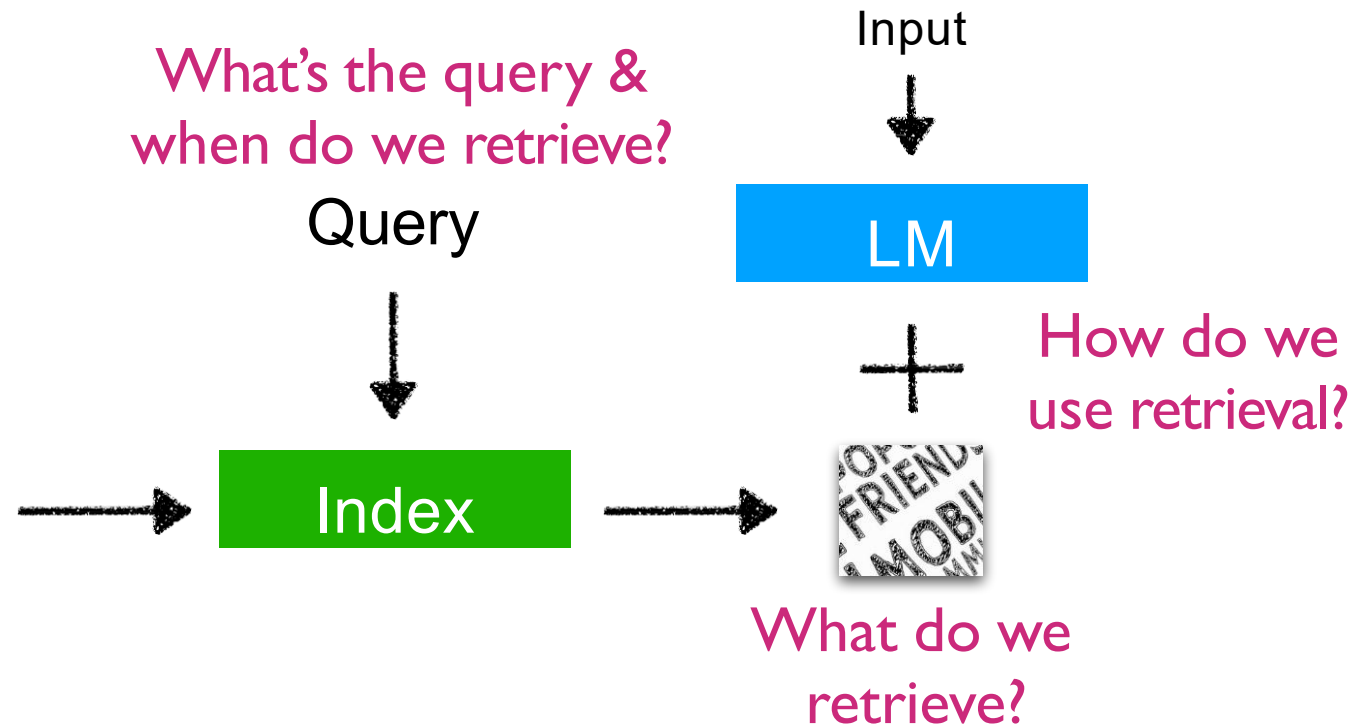
# Inference: Search



# Variations of RAG

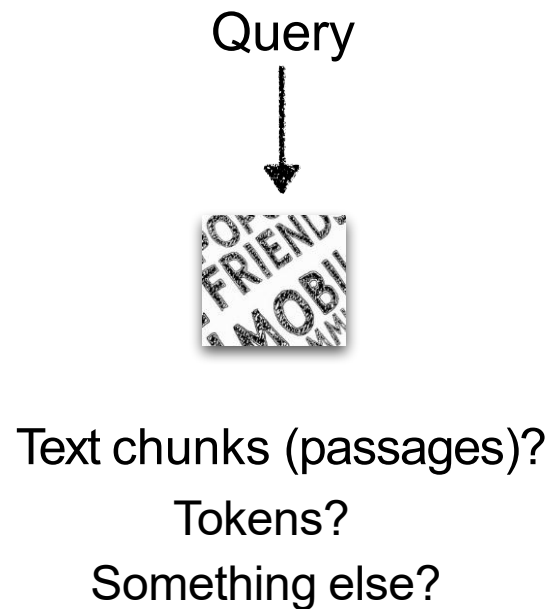


# Datastore

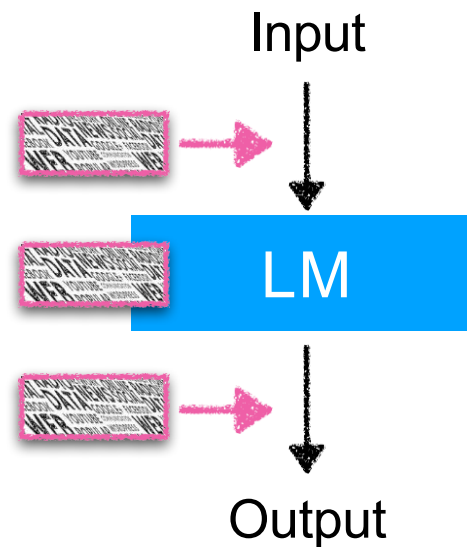


# Variations of RAG

**What** to retrieve?



**How** to use retrieval?



**When** to retrieve?



# In-Class Activity

This is also an exercise for reading academic papers to look for specific details

Each person in your group has a different paper.

Skim the paper given to you to just find the answers to these 3 questions.

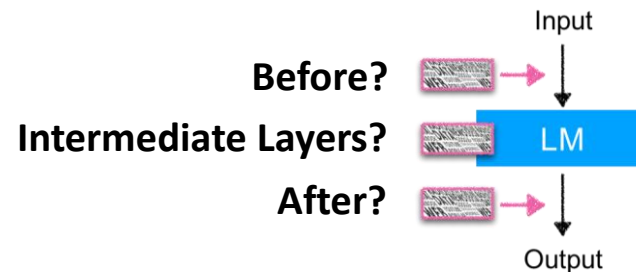
***What level of information do they retrieve?***

Query

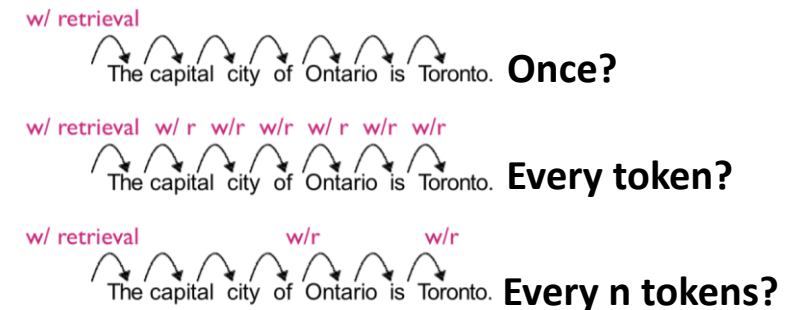


Text chunks (passages)?  
Tokens?  
Something else?

***How do they use retrieval with the model?***



***When do they retrieve?***



Share what you learn with your group.  
Submit on Blackboard when you're done.

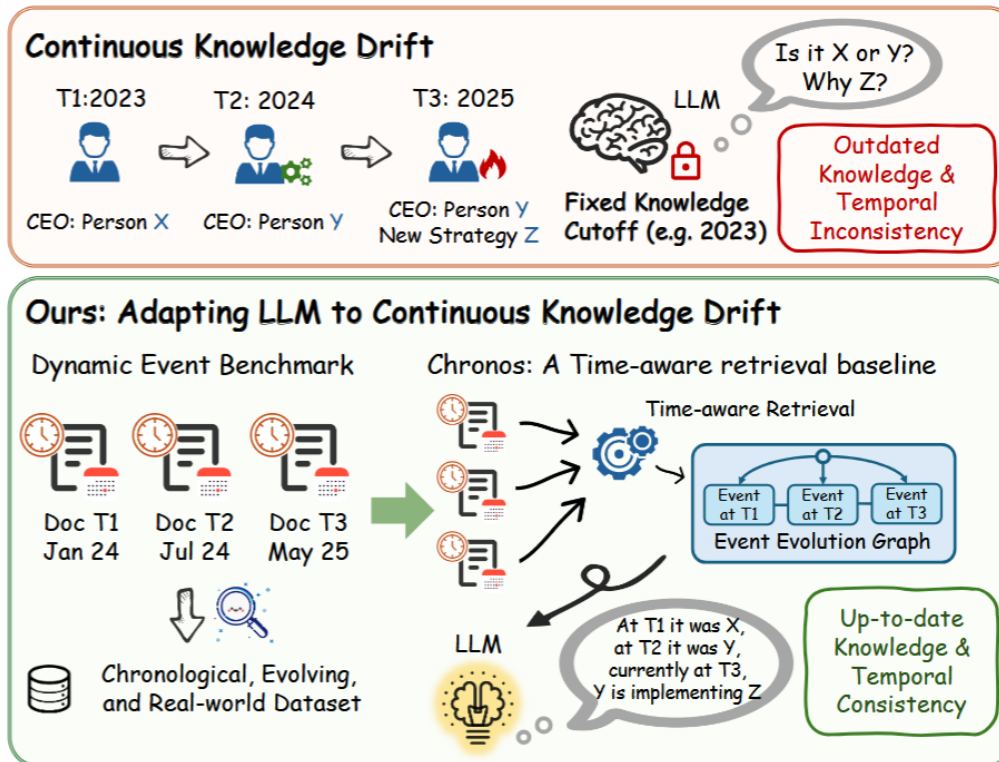
# Answers

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|                                 | What do retrieve? | How to use retrieval?                | When to retrieve?            |
|---------------------------------|-------------------|--------------------------------------|------------------------------|
| REALM (Guu et al 2020)          | Text chunks       | Input layer                          | Once                         |
| RETRO (Borgeaud et al. 2022)    | Text chunks       | Intermediate layers                  | Every n tokens               |
| kNN-LM (Khandelwal et al. 2020) | Tokens            | Output layer                         | Every token                  |
| FLARE (Jiang et al. 2023)       | Text chunks       | Output layer and then<br>Input layer | Every n tokens<br>(adaptive) |

*All models retrieve from the external text*

# Adapting to Knowledge Drift



## Knowledge

<World's Richest Person, held by, Elon Musk, 2024-01-01>  
<World's Richest Person, held by, Bernard Arnault, 2024-01-28>  
<World's Richest Person, held by, Jeff Bezos, 2024-03-05>  
<World's Richest Person, held by, Bernard Arnault, 2024-04-02>  
<World's Richest Person, held by, Elon Musk, 2024-06-08>  
<World's Richest Person, held by, Larry Ellison, 2025-09-10>  
<World's Richest Person, held by, Elon Musk, 2025-09-11>  
<Oracle stock price, surged to, USD 328, 2025-09-10>

## Historical Question

Who was the richest person in the world in March 2014?

## Contemporary Questions

C1: Who was the world's richest person on August 20, 2025?  
C2: Who were the world's richest people at any point during 2024?  
C3: Which company's stock surge led to Elon Musk losing his position as the world's richest person on September 10, 2025?

## Commonsense Question

What is the brightest star in the night sky?  
A. Sirius    B. Sun    C. Polaris    D. North Star

## Answers

Historical: Bill Gates    Commonsense: A    C1: Elon Musk  
C2: Elon Musk, Bernard Arnault, Jeff Bezos    C3: Oracle

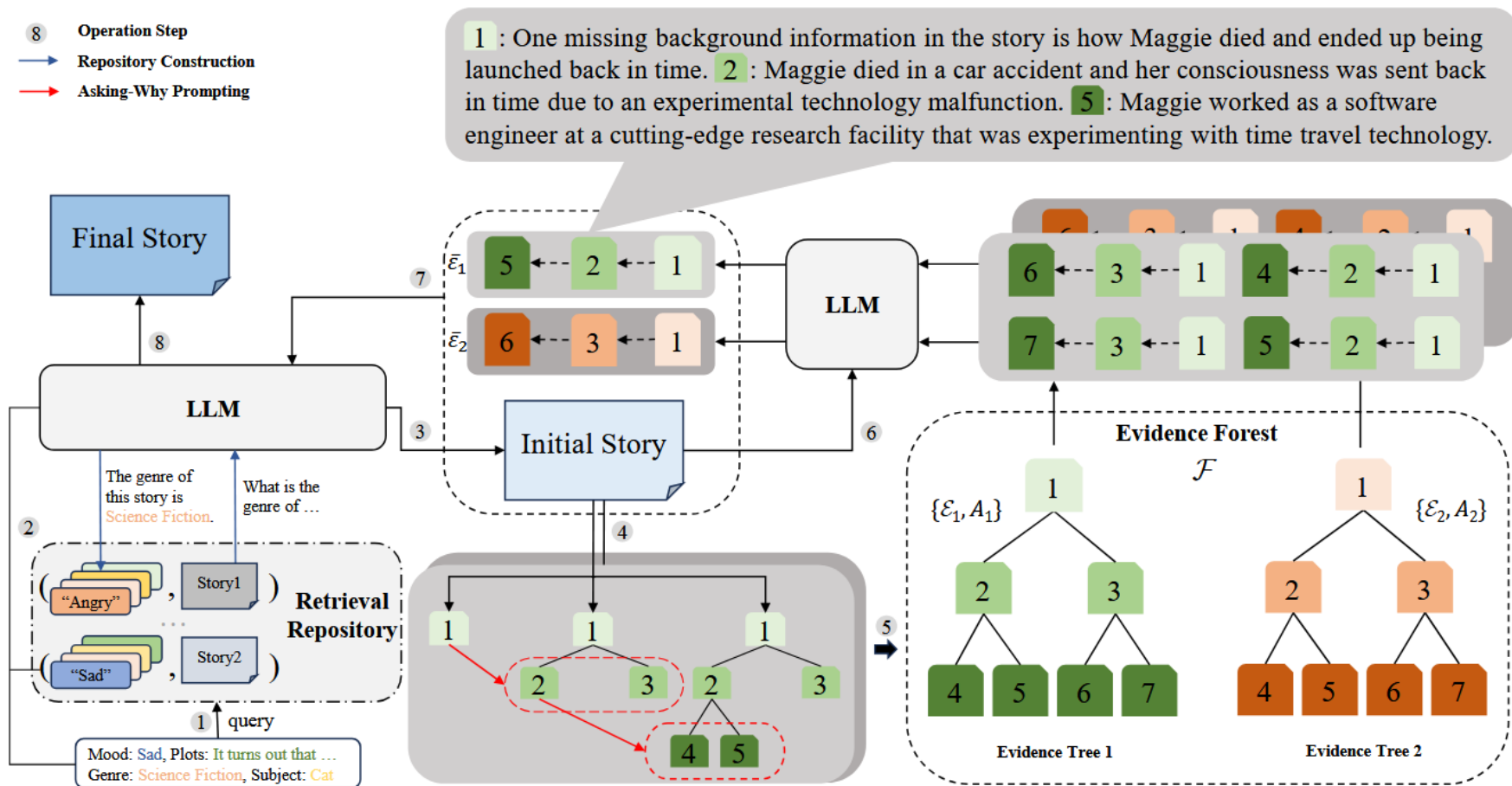
# Say Anything (2009)

Retrieval is not new

Before neural methods,  
retrieval was used for  
generating stories

| <i>User written sentence</i>                                                                              | <i>Matching sentence</i>                                                                                                                              | <i>Computer generated sentence</i>                                                                                |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| It all started easy enough.                                                                               | It all started out normal enough.                                                                                                                     | Got up, had lunch, got ready went to work etc.                                                                    |
| But then I noticed something very strange.                                                                | Then, as I flipped through the test, I realized something very strange.                                                                               | Mr. Sipiera gave me points on the blank page.                                                                     |
| Why would he do that? What was he up to?                                                                  | Why would he do that?                                                                                                                                 | Because he was planning to kill Sonya for her knowledge of the kidnapping plot.                                   |
| I decided to get to Sonya first and run away with her to a Greek island.                                  | We left pretty much after this and took a ferry back to Hong Kong island.                                                                             | Our original plan to go to Lamma island was ditched, as we had inadvertently spent half the day on Lantau island. |
| But we settled for Corfu, which has a beautiful little hotel on the edge of the old town.                 | But we settled for the simpler name.                                                                                                                  | We had dogs (and cats) when I was growing up but my wife and sons have never had a dog before.                    |
| Corfu was full of cats and dogs, and I ended up having to look at my feet a lot when walking around town. | and i ended up having to save the day.                                                                                                                | It's too long to list the details of the whole operation.                                                         |
| Sonya would take on the secret identity of a scuba instructor, and I'd be a fisherman.                    | "Before I even got so far as having a proper breakdown in realizing that fantasy is not reality, I found that I had the secret identity of a beetle." | "But that could have been very useful, right?"                                                                    |
| It meant we could spend a lot of secret time together on the jetty beyond the coastline.                  | She does spend a lot of time at the bar where they all hang out.                                                                                      | She has told me some of the men she hangs with have come on to her.                                               |
| I told her I'd kill them just like that old math teacher.                                                 | I told her I'd come help her clean and whatever else she needs.                                                                                       | Once she's gone, I'm pretty much on my own.                                                                       |
| Fishing all day, looking out for the police.                                                              | This is bad for the police.                                                                                                                           | If they stopped to survey the situation they could only see what was there -- none of the men were armed.         |
| It was a great secret life.                                                                               |                                                                                                                                                       |                                                                                                                   |

# GROVE (2023)



# Think-Pair-Share

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How might you evaluate a RAG system? What would you want to measure?